

Statistical Modelling of Grouped and Longitudinal Ecological Data: Survival Patterns in Moths and Wing-Span Growth Dynamics in Bats

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1 Introduction

Ecological studies explore how organisms interact with their environment, particularly with how factors can impact an organism's chances of survival and ability to grow. With the assistance of statistical techniques used to explore these associations, there is better alignment for dealing with complexity relating to both grouped or longitudinal data, allowing for inferences concluding an organism's biological outcome. The research conducted applied statistical techniques to an ecological (see this ... Bishop, 1972) study that collected data reporting *Biston betularia* (peppered moth) survival numbers, along with wing-span growth in *Pipistrellus pipistrellus* (*P.Pipistrellus*), one of Britain's most common bat species and a known predator of the peppered moth. Data from Ecological studies come in many forms, either in grouped or longitudinal format. In the context of this Study, peppered moths are nested within trees across different woodland areas, demonstrating a clear multilevel structure. In contrast, bats are recorded with repeated measurements captured for the growth of their wingspan over evenly spaced weekly time points. Having such dependencies introduces challenges with statistical modelling when accounting for variation across habitats or duration of time. Mixed-effects models allow hierarchical and temporal structures to be accounted for, random intercepts and slopes capture variability in outcomes between individuals (Bolker et al., 2009), and correlation structures accounting for dependence in repeated measures in longitudinal data (Fu et al., 2024). By fitting Mixed-effects models for both grouped and longitudinal data, we will evaluate the following outcomes for moths and bats separately:

Part I: Survival probability of peppered moths through City Centre Distance (Multilevel Data)

In areas of high industrial activity closer to the City Centre, pollution levels have impacted moths

through adapting their colour from a grey speckled colour to black. Data was captured for both sets of coloured moths, with both equal proportions of moth attached to trees across 24 woodland locations, each varying in distance (km) from the City Centre. Each tree accommodated 8 moths, with equal representation of black and grey moths (4:4). Dependencies like the moths' Gender (*Male/Female*) and Age Category (*Child/Juvenile/Adult*) were selected at random.

The aim of this data is to fit a model that helps evaluate whether the survival probabilities for black and grey-speckled moths changes as distance from the City Centre increases.

Part II: Bat Wing-Span Growth (Longitudinal Data)

A total of 32 common bats (*P.pipistrellus*) were captured, measured, tagged and returned to their respective roosting sites. Bats had their wingspan re-measured at successive 7-day intervals, accumulating to measurements up to 14 weeks total. The study looked to quantify the rate of growth in wingspan over time, and whether growth rate was associated with distance (km) from the City Centre.

Upon a final model being generated, we look to use our fitted model to answer the two following questions:

1. The probability that in week 10, bat 6 has a wingspan of under 68mm.
2. The probability of a bat roosting 9km from the City Centre and having a wingspan between 82-86mm at the time of the final measurement (Week 14).

After addressing both **Part I** and **Part II**, findings will be collated, and further discussion will be provided on the ecological processes for both bats and moths. Reference to the wider literature is also provided, referencing to how the findings from the study compare.

Survival Proportion of black/speckled moths

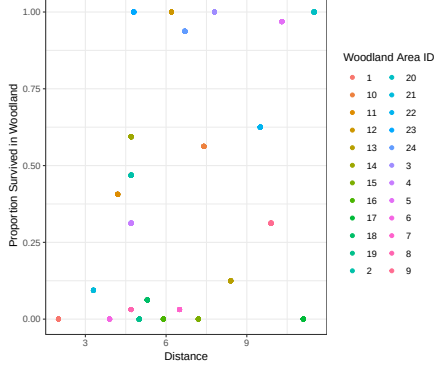


Figure 1: Proportion in Survival of all moths across the 24 woodland areas ranging in Distance, km, away from the City Centre.

2 Methodology

2.1 Part I: Moth Survival (Grouped / Multilevel Data)

2.1.1 Data Structure

The setup of the experiment explored Moth Survival (*Survived (1)/Died (0)*), both for dead black and speckled types, distributed with an equal ratio of 4 black and 4 speckled moths across trees situated in different woodland areas. 8 moths each were nested in a single tree, with 4 trees each located in 1 of the 24 woodland area situated between 2-11.5 kilometres (km) from a City Centre. Demographical and ecological covariates are supplemented with each moth, varying by the Moth's Age Category (*Infant/Juvenile/Adult*), Gender (*Female/Male*), morphing type (*Black, or Speckled*), and total distance (km) from the City Centre. Variation of survival among moths was evident across different Woodland areas, (Figure 1).

along with the trees they resided (Figure 2). Assessing survival of 8 moths (4 black/4 speckled) per tree, nested inside 1 of the 24 woodland areas, required the model to be fitted with random effects to account for group structures.

2.1.2 Model Specification

A binomial Generalized Linear Mixed Effects Model using a logit link function was required for predicting survival probability for *black* and *speckled* moths in trees across the 24 woodland areas.

$$\begin{aligned} Y_{ijk} &\sim \text{Bernoulli}(p_{ijk}) \\ \text{logit}(p_{ijk}) &= \beta_0 + \beta X_{ijk}^T + b_{0i} + c_{0j} + \epsilon_{ij} \\ b_{0i} &\sim N(0, \tau_{\text{woodland}}^2) \\ c_{0j} &\sim N(0, \tau_{\text{tree}}^2) \end{aligned}$$

Tree Survival Proportion of black/speckled moths by Woodland area

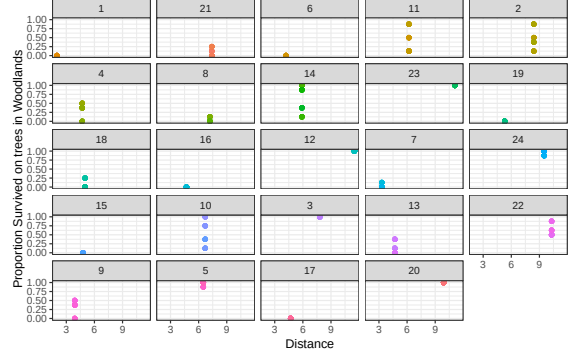


Figure 2: Proportion in Survival of all moths across each of the 4 trees situated in 1 of 24 woodland areas ranging in Distance, km, away from the City Centre.

Here, Y_{ijk} denotes the survival outcome for moth k on tree j within woodland i . The outcome variable is modelled as a Bernoulli random variable with success probability p_{ijk} . β_0 represents the intercept, the baseline log-odds for survival of a moth. βX_{ijk}^T holds all the effect sizes for the fixed terms *Distance* (km), *Morph* (black/speckled), *Age* (infant/juvenile/adult), and *Gender* (Male/Female) in the model. The notation extends to the following equation (below).

$$\begin{aligned} \beta X_{ijk}^T &= \beta_1 \cdot \text{Distance}_{ijk} + \beta_2 \cdot \text{Morph}_{ijk} \\ &+ \beta_3 \cdot \text{Age}_{ijk} + \beta_4 \cdot \text{Gender}_{ijk} \end{aligned}$$

The model uses all fixed effects provided in the model. Bootstrapping metrics were applied to justify the addition of random intercepts for both woodland i (b_{0i}) and tree j (c_{0j}) groups.

2.1.3 Model Diagnostics

AIC was used to determine the best model for predicting survival probabilities in both moth types, and model diagnostics are carried out to determine goodness of fit. Residuals produced through calculating the difference between observed and fitted observations were plotted against fitted values to assess normality assumptions. Additional plots include fixed and random effects against the residuals to examine if residuals fell between (-2,2) for 95% confidence intervals, satisfying normality conditions. These plots were for fixed effect *Distance* and each of the random effects b_{0i} and c_{0j} respectively. Random effects extended to QQ-plots to assess whether residuals follow the normal line.

Wingspan Growth of *P.pipistrellus* over weekly intervals (14 weeks Total)

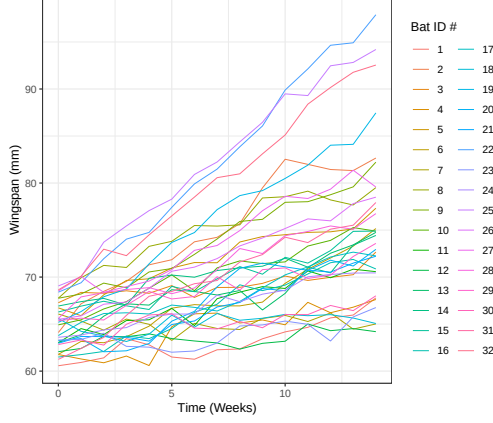


Figure 3: Weekly measurements for Wingspan growth in 32 bats of the *P.pipistrellus* species.

2.2 Part II: Bat Wingspan Growth (Longitudinal Data)

2.2.1 Data Structure

A total of 32 bats (*P.pipistrellus*) had weekly wingspan (*mm*) measurements, tracking each Bat's change in wingspan over the course of 14 weeks. The study aim was to predict wingspan over time, and evaluate if Distance (*km*) of their roosting site from the City Centre impacted wingspan growth.

Each bat, assigned a unique ID Number (*BatID*), differed in their initial wingspan (*Wingspan.mm*) at the start of the experiment. Baseline variation across the 32 bats therefore justified the inclusion of a random intercept (Figure 3). We also examine from the same figure that the gradient in growth patterns varies considerably, which led to the implementation of a random slope to account for individual wingspan growth rates over time.

2.2.2 Model Specification

Initial exploration of the data structure led to modelling a Generalized Linear Mixed-Effects model.

Evenly spaced weekly timepoints (*Time.week*) and distance from the City Centre (*Distance.km*) were modelled as fixed effects. Individual bats differed in their initial wingspans and growth rate trajectories (*Wingspan.mm*), leading to a random intercept and random slope for time (*Time.week*) being implemented to account for variation between bats (*BatID*). Justification for the random effect was reinforced through bootstrapping procedures. Temporal dependence in residuals were examined through visualizing an Autocorrelation Function (ACF), demonstrating the need for Auto-regressive

(AR) or auto-regressive moving average (ARMA) correlation structures to account for dependency between observations. The final model fitted for this longitudinal data on wingspan growth in bats is as follows:

$$Y_{it} \sim \mathcal{N}(\mu_{it}, \sigma^2)$$

$$\mu_{it} = \beta_0 + \beta_1 \text{Time}_{it} + \beta_2 \text{Distance}_i + b_{0i} + b_{1i} \text{Time}_{it} + \varepsilon_{it}$$

$$b_{0i} \sim \mathcal{N}(0, \tau_{\text{batID}}^2)$$

$$b_{1i} \sim \mathcal{N}(0, \tau_{\text{BatID}}^2)$$

$$\varepsilon_{it} = \rho \varepsilon_{i,t-1} + u_{it}, u_{it} \sim \mathcal{N}(0, \sigma_u^2)$$

Y_{it} represents the wingspan measurement for bat i at week t . Fixed effects of Time (*weeks*) and Distance (*km*) from the City Centre are included in the model. b_{0i} denotes the random intercept for each bat, accounting for individual wingspan measurements at baseline. $b_{1i} \text{Time}_{it}$ provides a random slope to model the rate of growth specific to each bat. ε_{it} represents a parametric autoregression (AR(1)) to model temporal dependence from repeated measurements for bat i over time. This dependence accounts for correlation of observations taken between weekly timepoints.

2.2.3 Model Diagnostics

Model adequacy for wingspan growth was assessed using several diagnostics.

Residual-fitted plots examined linearity, while QQ-plots for both residuals and random effects were used to assess normality within-bat wingspan variability. Autocorrelation was examined through ACF plots, guiding selection of the appropriate correlation structure (AR(1), ARMA etc.). Several correlation structures were examined, but AR(1) proved to be sufficient. The best models were compared using AIC to ensure the best final model that provided sufficient balance of fit and interpretability.

3 Data analysis

3.1 Part I: Moth Survival

Exploratory analysis examined significant differences in survival outcomes between *black/speckled* moths. The distribution of *died* vs. *survived* between the two moth types (Figure 4) showed no evidence of significant differences. Attention then turned to Figures 1 and 2 to establish whether random intercepts were needed. Parametric bootstrapping procedures confirmed that both intercepts were statistically significance, supporting their inclusion in the final model (Figure 5, 7).

Fixed Effect	exp[β (95% CI)]	SE ϵ	p-value
Intercept (β_0)	$3.36e^{-6}$ ($7.75e^{-9}$, $1.46e^{-3}$)	3.0987	<.001
Distance (β_1)	7.40 (3.12, 17.55)	0.4404	<.001
Morph (β_2)			
Speckled	0.62 (0.37, 1.14)	0.3103	0.123
Age (β_3)			
Juvenile	0.06 (0.02, 0.16)	0.4765	<.001
Adult	0.24 (0.11, 0.54)	0.4051	<.001
Gender:Male (β_4)	0.74 (0.39, 1.40)	0.3271	0.355
Random Effect	Variance (τ^2)		
Woodland ID (b_{0i})	13.156		
Tree ID (c_{0i})	3.648		
Model Diagnostics			
AIC	406.6		
log-Likelihood	-195.3		

Table 1: Binomial Generalized Linear Mixed-Effects Model

Quantifying the survival of moths, a summary of the best binomial GLMM is provided (Table 1). Our priority was to assess whether distance (in km) from the City Centre impacted the two morph types. Each unit increase in distance (km) led to the odds of survival increasing 7.4-fold, a strong positive effect. Speckled moths reported a 38% lower odds of survival compared to Black moths, but was not shown to be statistically significant ($p=0.123$). Compared to infant moths, all effects held constant, both Juvenile and Adult moths reported statistical significance for lower odds of survival. Gender was also included, but no statistical significance was present ($p=0.355$), with the model reporting 26% lower odds of survival. After parametric bootstrapping confirmed the inclusion of random intercepts for Woodland and Tree IDs, survival probabilities showed large variation. Accounting for all fixed effects, some woodlands proved to be more safer or riskier than others ($\tau^2=13.156$). Additional variation was present between trees, but the reported magnitude was smaller ($\tau^2=3.648$). Model diagnostics provided the best reported AIC (**406.6**) and log-likelihood (**-195.3**) from the analysis (Table 1). Examining the model diagnostics (Figure 8) for model adequacy, the residuals-fitted plot displayed a curved pattern, indicating potential non-linearity from the fixed terms provided. In contrast, plotting the residuals against distance alone, a random scatter demonstrates that the effect of distance was captured appropriately by the model. Diagnostics for woodland and tree random intercepts were examined in Figure 9, and both followed an approximate normal distribution. The random intercept for tree ID however, looked to deviate from the normal line for extreme values only.

3.2 Bat Wing-span Growth

Our GLMM implemented Time (weeks) and Distance (km) as fixed effects, and exploratory plots

(Figure 3) proved that random effects were needed to account for growth variability between bats. Parametric bootstrap was carried out to confirm the need for a random intercept on *bat ID*, and results from Figure 10 provided evidence for its inclusion. Exploratory analysis also demonstrated the difference in slopes between bats over time, and a random slope for individual bats over time (*Time.week*) was therefore included.

Fixed Effect	β (95% CI)	SE ϵ	p-value
Intercept (β_0)	62.85 (61.66, 64.05)	0.6117	<.001
Time, weeks (β_1)	0.78 (0.60, 0.95)	0.089	<.001
Distance, km (β_2)	0.45 (0.21, 0.68)	0.119	<.001
Random Effect	Variance (τ^2)		
Bat ID (b_{0i})	1.829		
Time/Bat ID ($b_{1i}Time_{it}$)	0.488		
Correlation Structure	Parameter Estimates (ρ)		
AR(1)	0.6341		
Model Diagnostics			
AIC	1470.967		
log-Likelihood	-727.4835		

Table 2: Generalized Linear Mixed-Effects Model predicting Wing-span growth in bats

With reference to Table 2, the average bat initially has a 62.85mm wingspan. Both *time (weeks)* and *distance (km)* proved statistically significant after holding all effects constant. Each additional week led to an increase of 0.78mm (0.60-0.95mm, $p<.001$), whereas each increase in distance by 1 km grew wingspan by 0.45mm (0.21-0.68mm, $p<.001$). Random effects that cover individual-bat variation includes a random intercept, and a random slope against *time*. The variability in wingspan between bats differs by a substantial 1.829mm, and the trajectory of a bat's wingspan growth can vary as much as 0.488 mm per 1 week increase. After accounting for random effects, autocorrelation function (ACF) was used to examine temporal dependence from the data. Figure 11 shows variation for several bars extending past the confidence bands, demonstrating statistical significance for temporal dependence. Model comparison using AIC found the First-Order Autoregressive (**AR(1)**) structure to be sufficient in accounting for the decay in autocorrelation (Figure 12).

The residuals-fitted plot for the longitudinal data on wingspan growth in bats showed a random scatter (Figure 13), indicating a constant variance. QQ-plots for both the random intercept and slope (Figure 14) closely followed the normal distribution line, with mild deviations at the extreme values for the random slope against *time*. This indicates that the random-effects in the GLMM satisfy a normal distribution, and that individual differences in bat growth trajectories was captured correctly.

After checking for model adequacy, we looked to use the model to predict the estimated wingspan of bats depending on the weekly timepoints and distance from City Centre.

The calculations were as follows:

"The probability that in week 10, bat 6 has a wingspan of under 68mm."

$$P(Y_{6,10} < 68) = \Phi\left(\frac{68 - \hat{y}_{6,10}}{\sigma_{\text{week } 10}}\right) \approx \mathbf{0.3275 \text{ (32.8\%)}}$$

Bat 6 has a 32.8% probability of having a wingspan under 68 mm by week 10 of the experiment (Figure 6).

"The probability of a bat roosting 9km from the City Centre and having a wingspan between 82-86mm at the time of the final measurement (Week 14)"

$$P(82 < Y_{i,14} < 86) = \Phi\left(\frac{86 - \hat{y}_{i,14}}{\sigma_{\text{week } 14}}\right) - \Phi\left(\frac{82 - \hat{y}_{i,14}}{\sigma_{\text{week } 14}}\right) \approx \mathbf{0.1528 \text{ (15.3\%)}}$$

The average bat at a roosting site 9km from a City Centre has a 15.3% probability of having a wingspan between 82-86mm by week 14 (Figure 15).

4 Discussion

4.1 Moth Survival

Grouped data was provided for dead *black and speckled* moths on their survival probability across distinct trees in woodland areas. Distance from the City Centre was the strongest predictor for greater chances of survival, which is supported by wider research on the declining populations of moth species in areas with low nitrogen and biodiversity (Fox et al., 2014). This is reflected by the almost zero coefficient for the intercept term (β_0). Outside of distance from the City Centre, there was significant reductions in survival in Juveniles and Adult moths compared to Infants. Otherwise, morph type (*black/speckled*) and gender reported non-significance in survival outcomes. The experiment however only utilizes dead moths, which doesn't account for behaviour dynamics between age groups which play a vital role in their ability to survive. Extended research looked to verify if *black* moths benefitted from advantages in protection from predators through being better camouflaged (Cook, 2013). Exploratory analysis examined survival rates in black moths closer to City Centres compared to speckled moths, but the analysis (Figure 4) had not demonstrated statis-

tical significance. An interaction term between distance and morph type was considered during model selection, but a lack of convergence prevented the validity of such findings from our data.

4.2 Bat Growth

Wingspan growth in *P. pipistrellus* was examined over-time using distance and Time (weeks) as fixed effects. Time was significantly associated with steady wingspan growth. In addition, distance from the City Centre was associated with higher wingspan growth, which can be ecologically explained through greater prevalence of foraging sites and fewer habitats in urban areas for bat species to thrive (Gili et al., 2020). Exploratory analysis confirmed different initial wingspans for *P. pipistrellus*, followed by varied growth rates over time. Research conducted on other bat species summarized how cooler and wetter climates along with insect abundance can both impact size at birth and post-natal growth (Eastick et al., 2022). Implementing an autoregressive (AR(1)) correlation structure reported high autocorrelation ($\rho \approx 0.63$), indicating that repeated wingspan measurements for the same bat were not independent over time. The AR(1) structure therefore helped capture temporal dependence successfully (Figure 12), allowing for reliable probability based inferences for predicting wingspan growth.

5 Conclusion

Mixed-effects models were applied across two ecological systems to investigate survival in moths and wingspan growth in bats. The analysis was able to conclude that survival for moths improved away from the City Centre, but that limitations with the sample size for model fitting prevented the ability to quantify if one morph type demonstrated greater odds of survival.

Further analysis for wingspan growth in bats revealed that roosting sites further away from the City Centre led to larger wingspan size over time. This effect was determined after accounting for both temporal growth trends and individual-level variation, reinforcing that environmental conditions in rural settings are better suited for bats. The data however doesn't account for such environmental characteristics such as noise pollution or the abundance of insects like the *black* and *speckled* moths. Further information on their environments can help us determine the ideal conditions needed for preserving future populations of *P. pipistrellus*.

References

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A Appendix

Boxplot of Died/Survived Moths by Morph Type

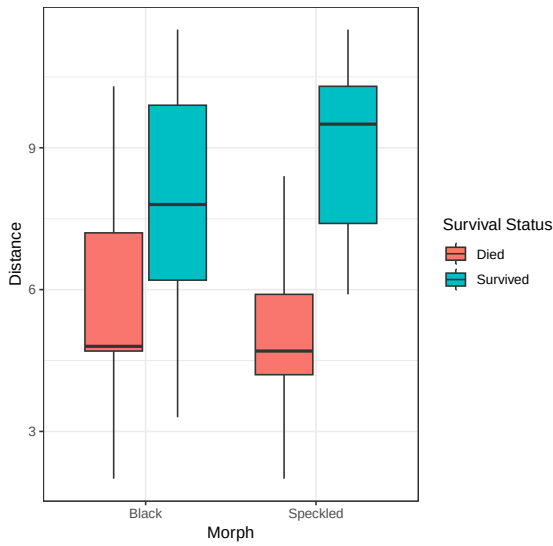


Figure 4: Caption Distribution of Black and Speckled dead Moths who'd died or survived the end of the experiment.

Parametric Bootstrap (*simulations = 100*)

$$H_0 : \tau_{\text{woodland}}^2 = 0$$

$$H_1 : \tau_{\text{woodland}}^2 > 0$$

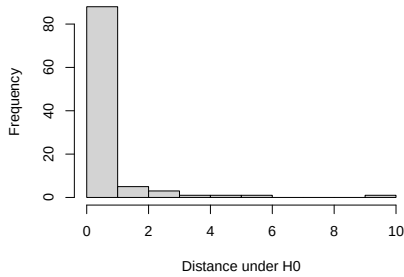


Figure 5: Parametric Bootstrap Distribution for the Woodland Random intercept Variance; Observed test statistic = **221.3246**, far outside the bootstrap distribution ($p < .001$)

$$\begin{aligned} \hat{y}_{6,10} &= 62.85 + (0.777)10 + (0.448)2.4 \\ &\quad + (-2.04) + (0.076)10 \\ &= \mathbf{70.41567} \end{aligned}$$

$$\begin{aligned} \text{Var}(Y_{6,10}) &= \sigma_{\varepsilon}^2 + \sigma_{b_0}^2 + (10^2)\sigma_{b_1}^2 \\ &\quad + 2(10)\text{Cov}(b_0, b_1) \\ &= \mathbf{29.23238} \end{aligned}$$

$$\sigma_{\text{week } 10} = \sqrt{\text{Var}(Y_{6,10})} = \mathbf{5.4066}$$

Figure 6: Parameters for evaluating "The probability that in week 10, bat 6 has a wingspan of under 68mm."

Parametric Bootstrap (*simulations = 100*)

$$H_0 : \tau_{\text{tree}}^2 = 0$$

$$H_1 : \tau_{\text{tree}}^2 > 0$$

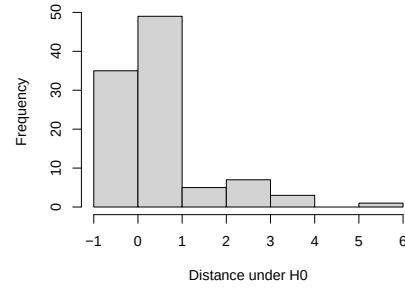


Figure 7: Parametric Bootstrap Distribution for the Woodland Random intercept Variance; Observed test statistic = **77.16025**, far outside the bootstrap distribution ($p < .001$)

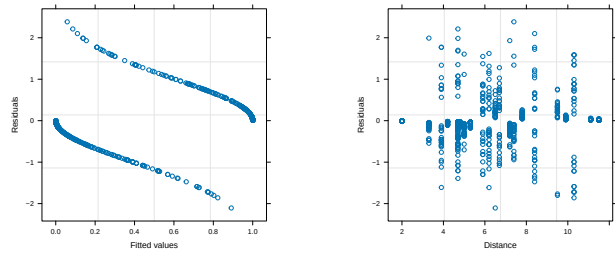


Figure 8: Moths Binomial GLMM Diagnostics; Residuals vs. Fitted (*left*); Residuals vs. Distance, in km (*right*);

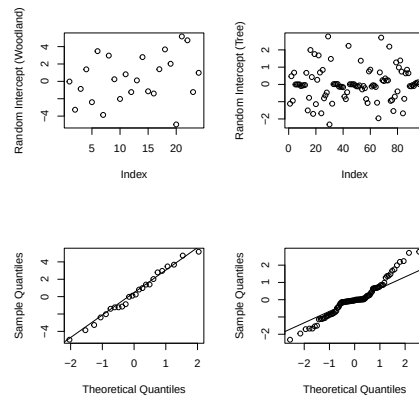


Figure 9: Random Intercept coefficients for Woodland (*top left*) and Tree (*top right*) groups, displaying the deviations from the overall intercept in the logit scale; Normal Q-Q plots for random effects, illustrating woodland (*bottom left*) and tree (*bottom right*) groups against a normal distribution

Parametric Bootstrap (*simulations = 1000*)

$$H_0 : \tau_{\text{batID}}^2 = 0$$

$$H_1 : \tau_{\text{batID}}^2 > 0$$

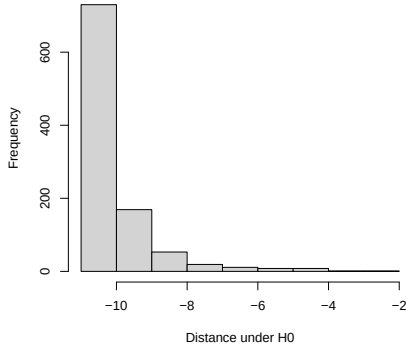


Figure 10: Parametric Bootstrap Distribution for the Bat ID Random intercept Variance; Observed test statistic = **1028.417**, far outside the bootstrap distribution ($p < .001$)

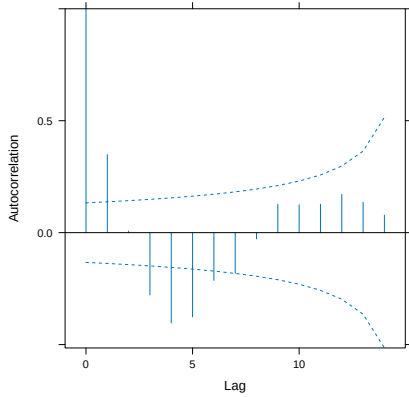


Figure 11: Autocorrelation plot prior to fitting a parametric model; Variation in peaks of autocorrelation surpassing outside the confidence bands implies the need for temporal dependence

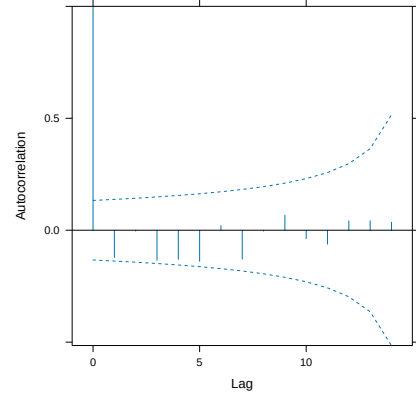


Figure 12: Autocorrelation plot after fitting a First-Order Autoregressive Correlation Structure; Lags fall within the confidence bands, with autocorrelation decaying towards zero

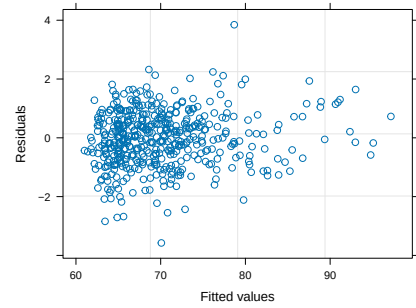


Figure 13: Fitted Values against residuals for the GLMM; Assessing normality through examining a random scatter, indicating constant variance across the Fitted values produced by the model

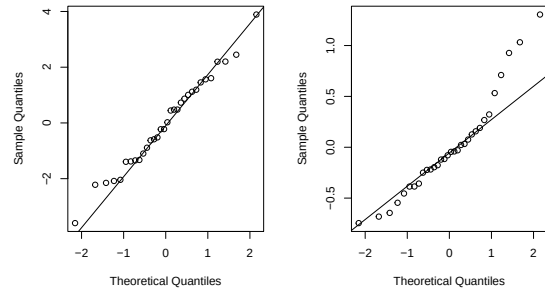


Figure 14: QQ-plots for BAT ID Random Intercept (*left*) and Bat ID slope with Time in weeks (*right*)

$$\hat{y}_{i,14} = 62.85 + (0.777) \cdot 14 + (0.448)9 = \mathbf{77.7629}$$

$$\begin{aligned} \text{Var}(Y_{i,14}) &= \sigma_{\varepsilon}^2 + \sigma_{b_0}^2 + (14^2)\sigma_{b_1}^2 \\ &\quad + 2(14) \text{Cov}(b_0, b_1) \\ &= \mathbf{52.38107} \end{aligned}$$

$$\sigma_{\text{week}14} = \sqrt{\text{Var}(Y_{i,14})} = \mathbf{7.23748}$$

Figure 15: Parameters for evaluating "The probability of a bat roosting 9km from the city centre has a wingspan of 82-86mm at the time of the final experiment."